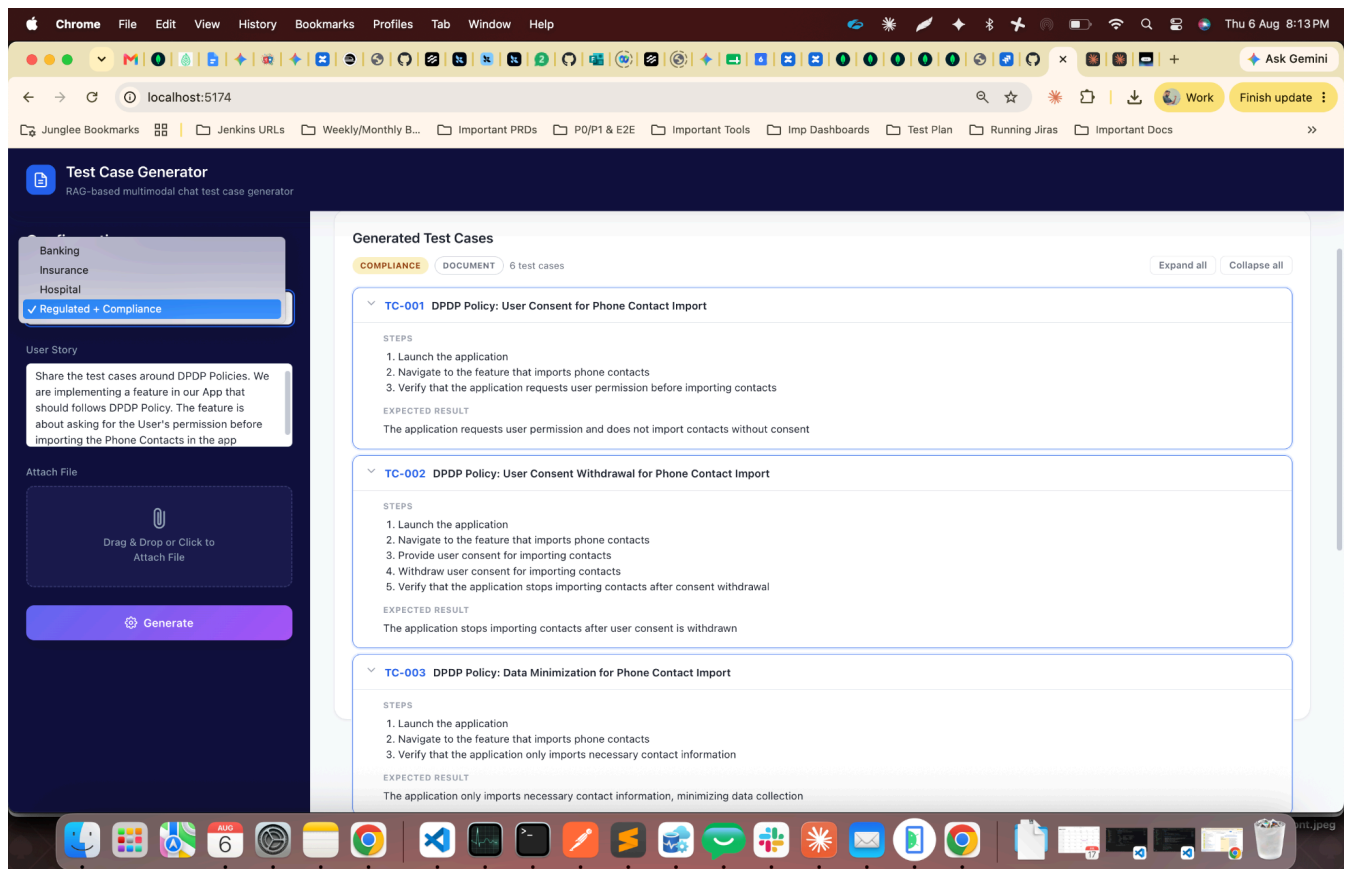
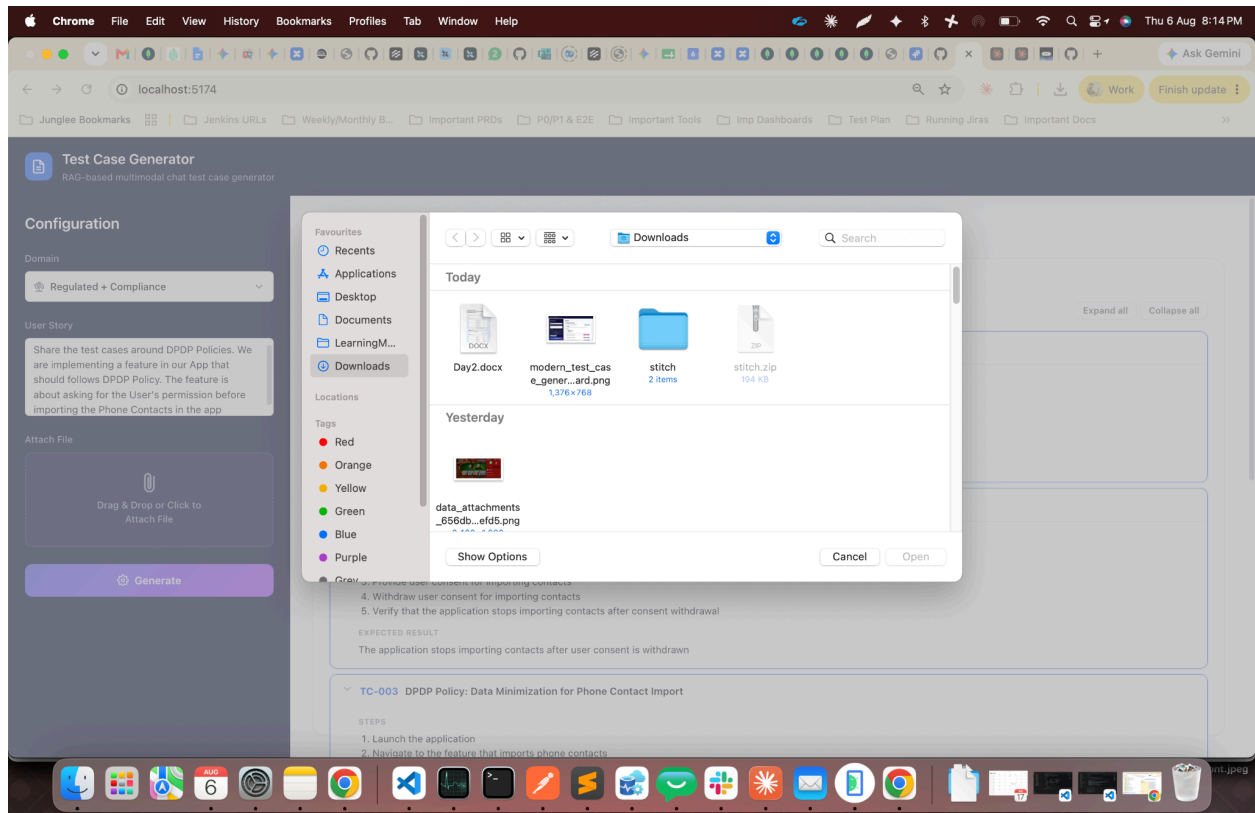


Home Assignment - Week 9 - Day 1& Day 2

UI Interface of the Web Application





Summary of the Changes:

1. Redesigned the Frontend (User Interface)

First round of changes:

I looked at the existing chat-style interface (built with React) and changed it from a single centered chat box into a two-part layout — a sidebar on the left and a main chat area on the right, similar to how Claude or ChatGPT looks. I added:

- A sidebar with the app's branding, a way to switch between domains (Banking/Insurance/Hospital), and a light/dark mode toggle that remembers your preference.
- Cleaner icons instead of plain text symbols.
- A responsive design — the sidebar automatically moves to the top of the screen on smaller screens/mobile.
- No new external tools were added; I just extended the existing styling system already in place.

Second round — matching a new design mockup:

I was given a new reference design (dark navy header/sidebar, a form on one side, and results shown as expandable cards on the other side). This wasn't just a style change — it changed how the app works: instead of a back-and-forth chat conversation, users now fill out a form (pick domain, enter user story, optionally attach a file) and click "Generate" once to get results.

- Removed the old chat-based conversation flow entirely and connected the app to a simpler backend endpoint that just takes input and returns test cases in one go.
- Rebuilt the main components: a form panel for input, and a results panel that handles loading, errors, empty states, and showing the actual results.
- Cleaned up old, unused code that was left over from the previous chat-based version.

I tested all of this manually in the browser (light mode, drag-and-drop file upload, generating results, error handling, and mobile view) and also ran a type-checking tool to catch any code errors.

2. Added a 4th Category: "Regulated + Compliance"

Instead of just adding another industry (like Telecom or Retail), I added a broader category that applies compliance and regulatory concerns *across* all industries — since Banking, Insurance, and Hospital all deal with strict compliance rules in similar ways.

This new category covers things like:

- Data privacy and user consent
- Keeping accurate, tamper-proof records of who did what (audit trails)
- Making sure only the right people have access to sensitive actions
- Rules around how long data can be kept, and how it should be deleted when required
- Identity verification checks
- Regulatory reporting requirements
- What happens if there's a data breach

To make this work properly, I didn't just add it to the dropdown menu — I updated it everywhere the app needed it:

- Added the new category to the backend logic that generates the actual test case content and instructions for that category.
- Made sure the backend's data validation was also updated — otherwise, it would have silently rejected any test cases generated for this new category.
- Added the option to the frontend dropdown.
- Added a new icon and a distinct color for this category so it's easy to tell apart from the other three.

I tested this by confirming all four categories correctly, running the type-checker again, and taking a screenshot to confirm the new option and icon show up properly.

JavaScript

```
const compliancePrompt = ChatPromptTemplate.fromMessages([
  [
    "system",
```

`You are an expert QA engineer specializing in cross-industry regulatory compliance frameworks.

Generate comprehensive, professional test cases using proper compliance terminology, covering key domains such as:

- Data privacy and consent management (GDPR/CCPA frameworks)
- Audit logs and trail integrity validation
- Role-Based Access Control (RBAC) and segregation of duties
- Data retention policies and right-to-erasure workflows
- KYC and AML screening controls
- Regulatory reporting requirements and statutory filing timelines
- Security breach notification protocols
- Data protection standards and encryption controls

Ensure all test cases are derived strictly from the provided context, encompassing positive, negative, and edge case scenarios.`

]
]);